

Refined Memo

Data 6560 Sport Analytics
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Risk vs. Reward: Analyzing Aggressive vs. Conservative Decision-Making in Golf

This memo refines my project proposal into a more measurable and data driven analysis of risk versus reward decision making in golf. The project examines how a golfer's level of aggressiveness in shot selection affects scoring outcomes, focusing on scenarios such as going for the green on par 5s or attempting recovery shots from the rough versus laying up safely. The goal is to identify the measurable conditions under which taking a risk yields better performance outcomes compared to conservative play.

Problem Definition & Justification

Decision making in golf presents one of the most difficult tradeoffs in all of sports. Players constantly weigh potential reward against the likelihood of penalty, yet these judgments are often guided by intuition rather than empirical data. The central research question driving this analysis is: *How does shot aggressiveness influence expected strokes gained relative to conservative decisions, considering factors such as lie type, distance to green, and player skill level?* The results could inform smarter strategy decisions for players and coaches, optimize course management, and ultimately improve performance consistency at all levels of play.

KPI's

The key metrics for this study will include expected strokes gained (ΔSG) as the primary indicator of performance, and success rate of aggressive decisions as a secondary metric representing the percentage of aggressive shots that yield positive outcomes compared to conservative alternatives. The unit of analysis will be individual golf shots, with attention given to par 5 and recovery situations from PGA Tour events between 2019 and 2024. Each observation will include shot context variables such as distance to hole, lie type (fairway, rough, sand, etc.), and resulting proximity to the green.

Data

Data will be sourced from the PGA Tour's ShotLink dataset, supplemented by Kaggle and DataGolf data if necessary. These sources provide detailed shot level information, including coordinates, proximity values, and strokes gained data. I plan to obtain and preprocess these datasets by labeling each shot as either "aggressive" or "conservative" based on criteria such as distance to the green and lie condition. If direct API access to ShotLink proves restrictive, open access Kaggle datasets will provide a suitable substitute for analysis.

Literature

Prior research supports this direction. Mark Broadie's *Every Shot Counts* (2014) first introduced strokes gained as a way for quantifying decision quality in golf. Subsequent studies, including PGA Tour analytics reports (2020–2023) and DataGolf's performance modeling research, have demonstrated strong relationships between approach shot aggressiveness,

proximity to the hole, and scoring results. However, most of these studies focus on player skill differentials rather than strategic decision making under risk, leaving an opportunity for this project to contribute new insights.

Modelling

My exploratory data analysis will begin with descriptive statistics and visualization of shot outcomes across aggressiveness categories. Preprocessing steps will include data cleaning, encoding categorical variables (e.g., lie type), and standardizing distance measures. I plan to begin with multiple linear regression to quantify how aggressiveness, distance, and lie type predict strokes gained, followed by decision tree modeling to identify thresholds that favor aggressive versus conservative choices. Model performance will be assessed using prediction accuracy metrics.

Risks/Considerations

The primary risks for this project are limited data access through PGA ShotLink and potential bias toward professional level data, which may not represent amateur conditions. These risks will be mitigated by relying on publicly available datasets and maintaining transparency about the research's professional context.

Next Steps

In the next week, my focus will be on obtaining/cleaning the dataset, defining aggressiveness classification rules, and conducting initial exploratory analysis to verify data quality. This work will ensure the foundation is in place for more advanced modeling and visualization in the next checkpoint.

Summary

By quantitatively examining risk taking behavior in golf, this project aims to provide actionable, data driven insights into when players should attack versus when they should play safe. The findings can directly inform both coaching strategies and player decision making grounded in analytics rather than intuition